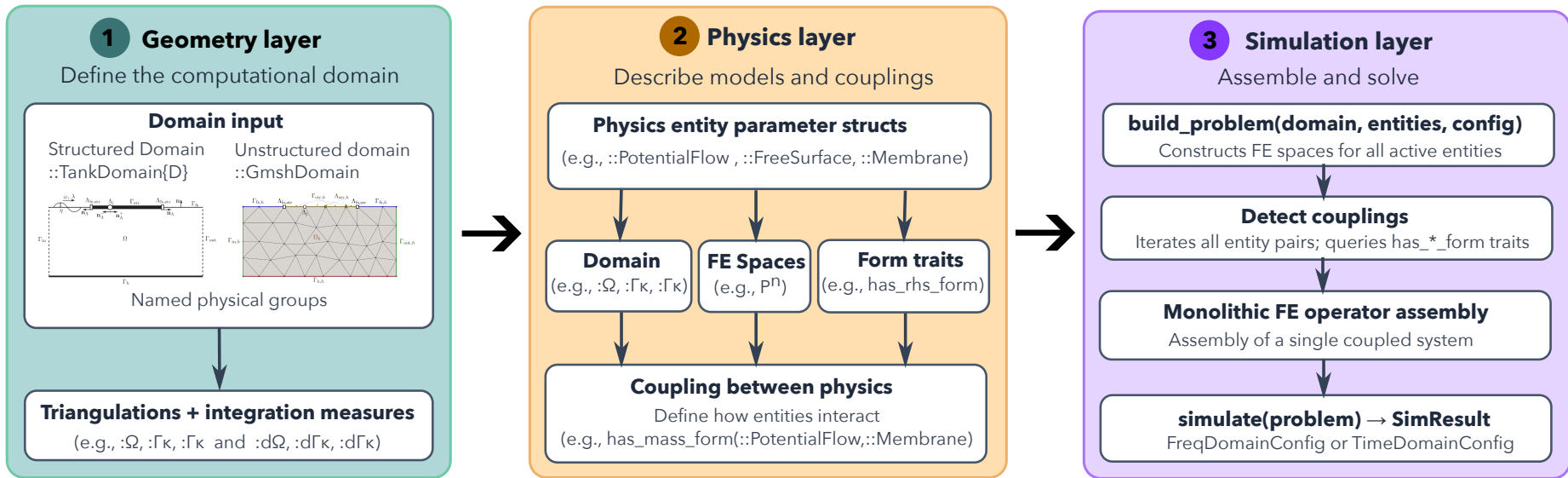


HydroElasticFEM.jl – Software Architecture



Novel Contributions

Monolithic FSI assembly

Fluid and structure solved in a single coupled linear system

Trait-discovered coupling

New entities activate coupling without editing assembly core

Unified freq / time API

Switch domain via config object; same physics entities throughout

SIPG for C¹ formulations

Beam/plate bending using standard Lagrange elements (no C¹ required)